

- Q.25 Draw and explain the flow sheet of a petroleum refinery.
- Q.26 Define and explain (i) Flash point (ii) Fire point.
- Q.27 Explain the significance of octane number and cetane number as quality parameters of petroleum fuels.
- Q.28 Write a note on viscosity and viscosity index of petroleum products and their importance for lubricating oils.
- Q.29 Explain Dubb's thermal cracking process with a neat diagram.
- Q.30 Describe the fluidized bed catalytic cracking (FCC) process and its advantages over fixed bed process.
- Q.31 Explain the chemical reactions involved in catalytic reforming of naphtha..
- Q.32 Describe the Thermofer catalytic Reforming (TCR) moving bed process with diagram.
- Q.33 Write a note on Reforming and its types.
- Q.34 Explain the MEK (methyl Ethyl Ketone) dewaxing process for lubricating oils.
- Q.35 Describe Visbreaking process with neat diagram.

#### SECTION-D

**Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Write about pre-treatment of crude oil and explain any one process with a neat sketch.
- Q.37 What is polymerization? Explain hot sulfuric acid polymerization process and draw neat, labelled pdiagram.
- Q.38 Write a comprehensive note on Purification of Petroleum Products. Describe the urea dewaxing process with a neat sketch.

(60)

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#### 5th Sem / Chem Sub : Petroleum Refining

Time : 3Hrs.

M.M. : 100

#### SECTION-A

**Note:** Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The origin of petroleum is best explained by which theory?
- Inorganic theory
  - Organic (biogenic) theory
  - Cosmic theory
  - Volcanic theory
- Q.2 Electrical desalting of crude oil is done to remove:
- Sulphur compounds
  - Water and dissolved salts
  - Light hydrocarbons
  - Wax
- Q.3 The boiling range of kerosene obtained from fractional distillation of crude oil is approximately:
- Below 70°C
  - 70-120°C
  - 150-250°C
  - Above 350°C
- Q.4 Which of the following is NOT a product of fractional distillation of crude oil?
- LPG
  - Bitumen
  - Polyethylene
  - Diesel

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- Q.5 Flash point is defined as the lowest temperature at which a petroleum product:
- Starts boiling
  - Gives off vapours that ignite momentarily on application of a flame
  - Solidifies
  - Changes colour
- Q.6 Cetane number is a measure of the ignition quality of:
- Petrol
  - Kerosene
  - Diesel fuel
  - LPG
- Q.7 Visbreaking is a mild form of:
- Catalytic cracking
  - Thermal cracking
  - Reforming
  - Polymerisation
- Q.8 The catalyst used in Houdry's fixed bed catalytic cracking process is:
- Zeolite
  - Platinum on alumina
  - Acid treated clay (silica-alumina)
  - Iron oxide
- Q.9 The primary purpose of catalytic reforming of naphtha is to:
- Reduce sulphur content
  - Increase octane number
  - Remove wax
  - Lower pour point
- Q.10 Doctor's sweetening process is used to remove
- Wax from lubricating oils
  - Asphalt from heavy oils
  - Mercaptans (thiols) from petroleum products
  - Water from crude oil

## SECTION-B

**Note:** Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Name any two Indian petroleum refineries with their locations.
- Q.12 Bitumen is obtained as the lightest fraction from the fractional distillation of crude oil. (True / False)
- Q.13 What is the boiling range of gasoline (petrol) from fractional distillation?
- Q.14 The process of removing emulsified water from crude oil is called \_\_\_\_\_.
- Q.15 What is the full form of FCC in petroleum refining?
- Q.16 Name any one type of reforming process.
- Q.17 The viscosity index of a lubricating oil decreases as the temperature increases. (True / False)
- Q.18 Name any one de waxing process used in petroleum purification.
- Q.19 Name the catalyst used in non-regenerative fixed bed process.
- Q.20 Unit of viscosity is \_\_\_\_\_.

## SECTION-C

**Note:** Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Describe the composition and classification of petroleum.
- Q.22 What problems does water in crude oil cause?
- Q.23 Write a note on the transportation of crude oil.
- Q.24 Describe the fractional distillation of crude oil. List the major fractions with their boiling ranges.